

Observation of High-Order Polarization-Locked Vector Solitons in a Fiber Laser

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We report on the experimental observation of a new type of polarization-locked vector soliton in a passively mode-locked fiber laser. The vector soliton is characterized by the fact that not only are the two orthogonally polarized soliton components phase-locked, but also one of the components has a double-humped intensity profile. Multiple phase-locked high-order vector solitons with identical soliton parameters and harmonic mode locking of the vector solitons were also obtained in the laser. Numerical simulations confirmed the existence of stable high-order vector solitons in the fiber laser.

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Solitons, as stable localized nonlinear waves, have been observed in various physical systems and have been extensively studied [1,2]. Optical solitons were first experimentally observed in single mode fibers (SMFs) by Mollenauer, Stolen, and Gordon in 1980 [3]. It was shown that the dynamics of the solitons could be well described by the nonlinear Schrödinger equation, a paradigm equation governing optical pulse propagation in ideal SMFs. However, in reality, a SMF always supports two orthogonal polarization modes. Taking fiber birefringence into account, it was later found that, depending on the strength of fiber birefringence, different types of vector solitons, such as the group velocity-locked vector solitons [4,5], the rotating polarization vector solitons [6,7], and the phase-locked vector solitons [8–10], could also be formed in SMFs.

Optical solitons were also observed in mode-locked fiber lasers. Pulse propagation in a fiber laser cavity is different from that in a SMF. Apart from propagating in the fibers that form the laser cavity, a pulse propagating in a laser is also subject to actions of the laser gain and other cavity components. The dynamics of solitons formed in a fiber laser are governed by the Ginzburg-Landau (GL) equation, which takes account not only of the fiber dispersion and Kerr nonlinearity but also the laser gain and losses. However, it was shown that, under suitable conditions, solitons formed in fiber lasers have analogous features to those of solitons formed in SMFs. Furthermore, vector solitons were also predicted in mode-locked fiber lasers and recently confirmed experimentally [11,12].

Among the various vector solitons formed in mode-locked fiber lasers or SMFs, the phase-locked one has attracted considerable attention. Back in 1988, Christodoulides and Joseph first theoretically predicted a form of phase-locked vector soliton in birefringent dispersive media [8], which is now known as a high-order phase-locked vector soliton in SMFs. The fundamental form of the phase-locked vector solitons was recently experimentally observed [12–14]. However, to the best of our knowledge, no high-order temporal vector solitons have been

demonstrated. Numerical studies have shown that the high-order phase-locked vector solitons are unstable in SMFs [7]. In this Letter, we report on the experimental observation of a stable phase-locked high-order vector soliton in a mode-locked fiber laser. Multiple high-order vector solitons with identical soliton parameters coexisting in a laser cavity and harmonic mode locking of the high-order vector solitons were also observed. Moreover, based on a coupled Ginzburg-Landau equation model, we show numerically that phase-locked high-order vector solitons are stable in mode-locked fiber lasers.

The experimental setup is shown in Fig. 1. The fiber laser has a ring cavity consisting of a 4.6 m length of erbium-doped fiber with group velocity dispersion parameter 10 ps/km/nm and a total length of 5.4 m of standard SMF with group velocity dispersion parameter 18 ps/km/nm. Mode locking of the laser is achieved with a semiconductor saturable absorption mirror (SESAM). A polarization-independent circulator was used to force the unidirectional operation of the ring and simultaneously to incorporate the SESAM in the cavity.

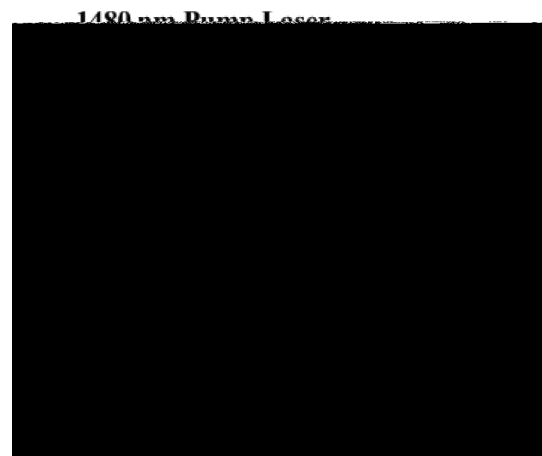


FIG. 1. Schematic of the vector soliton fiber laser. WDM: Wavelength division multiplexer. EDF: Erbium-doped fiber.

results show that the formation of high-order phase-locked vector solitons is a natural consequence of pulse circulation in the fiber laser under the current operation conditions.

To confirm our experimental observations, we also numerically simulated the operation of the laser. We used the following coupled Ginzburg-Landau equations to describe the pulse propagation in the weakly birefringent fibers in the cavity:

$$\begin{aligned} \frac{\partial u}{\partial z} = & \frac{1}{4} i \beta u - \delta \frac{\partial u}{\partial t} - \frac{i k^{00}}{2} \frac{\partial^2 u}{\partial t^2} + \frac{i k^{000}}{6} \frac{\partial^3 u}{\partial t^3} \\ & + i \gamma |u|^2 u + \frac{2}{3} j |v|^2 u + \frac{i \gamma}{3} v^2 u + \frac{g}{2} u + \frac{g}{2 \Omega_g^2} \frac{\partial^2 u}{\partial t^2}, \\ \frac{\partial v}{\partial z} = & \frac{1}{4} i \beta v + \delta \frac{\partial v}{\partial t} - \frac{i k^{00}}{2} \frac{\partial^2 v}{\partial t^2} + \frac{i k^{000}}{6} \frac{\partial^3 v}{\partial t^3} \\ & + i \gamma |v|^2 v + \frac{2}{3} j |u|^2 v + \frac{i \gamma}{3} u^2 v + \frac{\kappa}{3} v T_s T T_s T c \beta_s T T T T i v u T T T T T s n T \beta T T T v \end{aligned}$$

vector soliton is a single-hump soliton. It is interesting to see that the pulse of the strong component is only temporally overlapped with one of the two pulses of the weak component. Because of the strong cross-phase coupling between the temporally overlapped pulses, the two pulses of the weak components have different pulse widths and intensities. Propagating within the cavity, obvious coherent energy exchange between the two temporally overlapped solitons is visible. Figure 4(b) further gives the calculated spectra of the vector soliton components, which also show that the phase difference between the two bound solitons of the weak component is 180° .

Depending on the laser parameters selected, other high-order phase-locked vector solitons, such as the one with both soliton components having a double-humped structure, were also numerically obtained. We note that similar high-order vector solitons were also predicted for pulse propagation in weakly birefringent fibers, but they are unstable. However, we found that all of the high-order phase-locked vector solitons obtained numerically were stable in the laser. We believe that the differences in stability of the high-order phase-locked vector solitons between fiber and fiber lasers could be traced back to their different soliton natures. While the soliton formed in a SMF is essentially a Hamiltonian soliton, the one formed in a fiber laser is a dissipative soliton, which is in fact a strange attractor of the laser system. The formation of multiple identical high-order vector solitons in the fiber laser clearly shows the dissipative nature of the vector solitons formed.

In conclusion, we have reported the first experimental observation of a type of high-order phase-locked vector soliton in a passively mode-locked fiber laser. The high-order vector soliton is characterized by the fact that its two orthogonal polarization components are phase-locked, and, while the stronger polarization component is a single-hump pulse, the weaker component has a double-humped structure with 180° phase difference between the humps. Our experimental results are the first confirmation of the theoretical predictions of high-order phase-locked vector

solitons in birefringent dispersive media. Finally we note that, although the current experiment was conducted on a mode-locked fiber laser whose dynamics is governed by the coupled GL equations, due to the wide applicability of the coupled GL equations in describing the dynamics of real physical systems, it is expected that similar results could also be observed in other systems.

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